

Version: 1.0

SOFTWARE REQUIREMENTS (SRS)

School Finder Mobile Application

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Version: 1.0

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Estimated Budget: \$1,300 USD

The School Finder Mobile Application is a smart educational institution discovery system designed to help students search, compare, evaluate, and navigate schools and universities efficiently. The application integrates cloud technologies, mobile responsiveness, ratings systems, school comparison tools, administrative dashboards, and real-time educational information management.

1. Introduction

The School Finder Mobile Application was developed to solve the difficulties students face when searching for educational institutions. Many students struggle to obtain reliable information regarding tuition fees, academic programs, facilities, locations, and ratings of institutions. This project centralizes educational information into one digital platform accessible through Android smartphones. The application enables students to search institutions by category, location, tuition, ratings, and available programs. Parents can also use the platform to support educational planning for their children. Educational institutions benefit by showcasing their academic offerings and facilities to prospective students. The project integrates modern technologies such as Android Studio, Firebase Authentication, PostgreSQL databases, Cloudinary cloud storage, and Google Maps APIs. The system is scalable, secure, and optimized for educational discovery and comparison.

2. Objectives of the Project

The major objective of the School Finder application is to simplify educational institution discovery and improve accessibility to reliable academic information. Additional objectives include improving transparency through reviews and ratings, helping students compare institutions effectively, and enabling institutions to showcase educational services digitally. The system also aims to improve decision-making processes by providing detailed institution profiles, comparison workflows, and integrated maps for navigation.

3. Overall Description

The School Finder system operates using a client-server architecture where the Android mobile application communicates with backend REST APIs. School information is stored in PostgreSQL databases while Firebase Authentication secures user accounts and login sessions. Cloudinary services handle image uploads and school gallery management while Google Maps APIs provide real-time location tracking and navigation support. The application supports school search, ratings, reviews, favorites management, comparisons, and administrator analytics dashboards. The project is designed to remain responsive across multiple Android devices while maintaining optimized application performance and scalability.

4. User Roles & Responsibilities

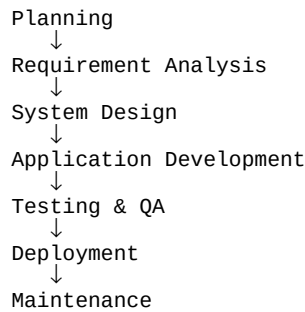
Role	Responsibilities
Guest User	Browse schools and search institutions
Registered Student	Rate schools, submit reviews, save favorites
Administrator	Manage schools, users, reports, and analytics

The permission structure ensures secure access control and role-based management of system operations.

5. System Architecture

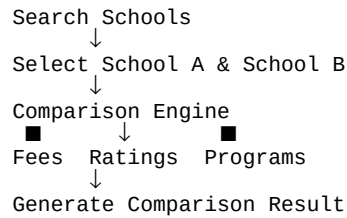
The School Finder application follows a three-tier architecture consisting of the frontend layer, backend layer, and database layer. The frontend layer is developed using Android Studio and Java/Kotlin programming languages. The backend layer uses Spring Boot REST APIs to process requests and communicate with the database. PostgreSQL databases manage school records, reviews, ratings, and user information. External integrations including Firebase Authentication, Cloudinary, and Google Maps APIs improve scalability, security, and user experience. The architecture ensures maintainability and future system expansion.

6. SDLC Workflow Diagram



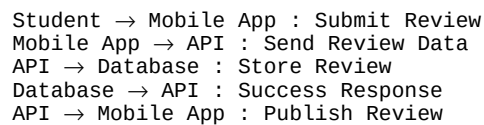
The SDLC process ensures proper planning, implementation, deployment, and long-term maintenance of the School Finder application.

7. School Comparison Workflow Diagram



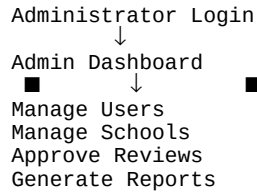
The comparison engine allows users to analyze schools side-by-side using tuition, ratings, facilities, programs, and location information.

8. Review & Rating Sequence Diagram



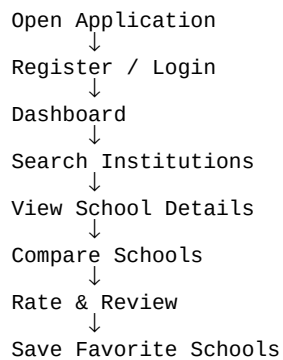
The ratings and reviews workflow promotes transparency and allows students to provide valuable educational feedback.

9. Admin Management Workflow



The administrative dashboard centralizes all management operations including user control, school moderation, review approvals, and analytics monitoring.

10. Enhanced User Flow Diagram



The enhanced user workflow demonstrates how students interact with the application from authentication to educational decision-making.

11. Functional Requirements

The School Finder Mobile Application supports several functionalities including authentication, school search systems, school comparison systems, reviews and ratings management, favorites management, cloud image hosting, GPS navigation, and administrator dashboards. Users can search institutions by tuition, ratings, location, category, and academic programs. Registered students can submit reviews and ratings while administrators manage user activities, schools, and reports.

12. Non-Functional Requirements

The application must remain secure, scalable, maintainable, and responsive. Pages should load quickly and support multiple Android devices and screen sizes. The system uses HTTPS encryption, Firebase Authentication, and secure cloud integrations to protect user information and system communications.

13. Database & APIs

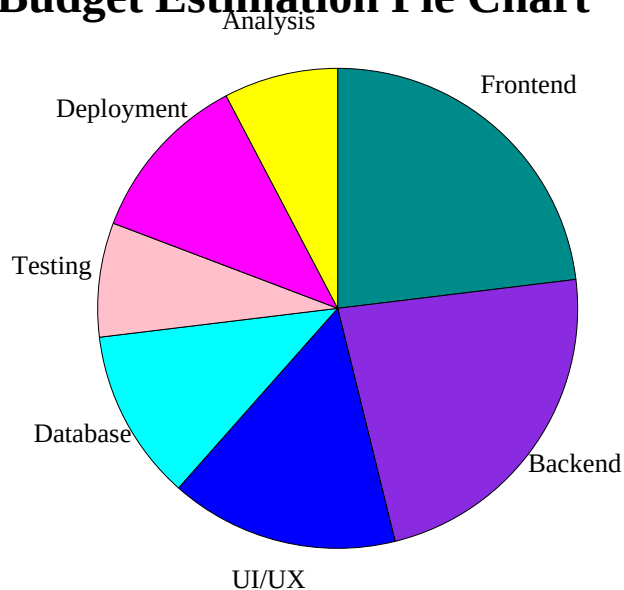
The system database stores users, schools, reviews, ratings, and favorites information. REST APIs manage communication between the Android application and backend systems. The APIs support authentication, school retrieval, review submissions, ratings management, and administrator operations.

14. Budget & Cost Analysis

The School Finder Mobile Application has an estimated implementation budget of \$1,300 USD. The project budget includes requirement analysis, UI/UX design, frontend development, backend APIs, database configuration, testing, deployment, and documentation. The project budget is carefully distributed to maintain cost efficiency while ensuring application quality and scalability.

Module	Estimated Cost
Requirement Analysis	\$100
UI/UX Design	\$200
Frontend Development	\$300
Backend Development	\$300
Database Design	\$150
Testing & QA	\$100
Deployment & Documentation	\$150

15. Budget Estimation Pie Chart



The pie chart illustrates the proportional distribution of the project implementation budget.

16. Recommendations

For Developers: Maintain secure coding standards, optimize application performance, and implement scalable architectures for future expansion.

For Users: Keep applications updated and provide honest educational reviews and ratings.

For Educational Institutions: Update school information regularly and provide accurate academic details and facilities information.

17. Conclusion

The School Finder Mobile Application provides a modern educational discovery platform that simplifies school comparison, educational research, ratings management, and institutional navigation. The project integrates Android technologies, cloud storage, authentication systems, and database management solutions to deliver a secure and scalable educational application. This Software Requirements Specification document provides a complete overview of the project objectives, architecture, workflows, diagrams, requirements, budget estimations, and implementation strategies required for successful project development and deployment.